

Screen to Sky (S2S) Backbone Report — v5

Build History

Version	Stories	Summary
v1	ST-511	Foundation: auth, path view, CSV upload, basic scoring, admin
v2	ST-802B/C/D	Ground school content, Quiz-Seed system, debrief reveal
v3	ST-802D+	L2 GLGL doctrine, go-around detection, Machado quiz gate
v4	ST-963	Engine v2: landing scorer, GTGL sequence detector, commitment gate, safety overrides, per-leg scoring
v5	ST-900	Admin gradebook, shadow mode (impersonation), path view overrides, INSTRUCTOR role

1. Architecture

Layer	Technology
Framework	App Router (server/client component separation)
Database	PostgreSQL via Prisma ORM
Auth	NextAuth.js v4, JWT strategy, Credentials provider
Storage	S3 via presigned URLs (telemetry CSVs)
Visualization	Recharts (debrief charts)
Animation	Framer Motion
Scoring Engine	Custom TypeScript (lib/s2s-engine/)

2. Database Schema

```

User (id, email, password_hash, role[ADMIN|INSTRUCTOR|STUDENT], name)
  [ ] Account (NextAuth)
  [ ] AuthSession (NextAuth)
  [ ] FlightSession[] (telemetry uploads)
  [ ] UserProgress[] (per-lesson tracking)
  [ ] QuizAttempt[] (quiz-seed-based attempts)

Lesson (id, lesson_num[1-17], arc_num[1-5], title, description, ...)
  [ ] objectives: JSON string[]
  [ ] Readiness flags: flightSchoolReady, groundSchoolReady, quizReady, scoringReady, mediaReady
  [ ] Production notes: 4 text fields for content agents
  [ ] FlightSession[]
  [ ] UserProgress[]

Quiz (id, quiz_num[250-253], title, questions: JSON, required_score)
  [ ] Legacy check-flight quizzes only

GroundSchoolContent (id, lesson_num[@unique], title, arc, mode, ...)
  [ ] acsAlignment, repWalkthroughFocus, beforeYouFly: JSON
  [ ] mission, whatMattersToday, targetNumbers: JSON
  [ ] simpleFlightFlow, commonMistakes, doNotWorryAboutYet: JSON
  [ ] debriefReveal: JSON
  [ ] cfiNotes: String? (nullable)

QuizSeed (id, lesson_num, question, options: JSON, correct_index, explanation)
  [ ] 5 seeds per lesson (L1-L4 seeded, 4/5 to pass)

QuizAttempt (id, user_id, lesson_num, answers: JSON, correct_count, total_count, passed)

FlightSession (id, user_id, lesson_id, csv_filename, cloud_storage_path, parsed_data: JSON)
  [ ] Attempt[] (flight segments)
  [ ] Score[] (1-5 scale)

Attempt (id, session_id, segment_type, start_time, end_time, metrics: JSON, confidence)

Score (id, session_id, overall_score[1-5], breakdown: JSON)

UserProgress (id, user_id, lesson_id, flight_score, quiz_passed, debrief_viewed, status)
  [ ] Unique constraint: [userId, lessonId]

```

3. S2S Scoring Engine (lib/s2s-engine/)

Pure TypeScript engine — no Python dependency. Runs in Node.js production.

Core Pipeline

CSV Text → loadTelemetry(csv, 1Hz) → detectPhases(df) → gradeL#(df, phases) → Lesson-Result



toDict() / toHtml()

Module Inventory

Module	File	Purpose
Entry Point	<code>index.ts</code>	<code>evaluateFlight(csvText, lessonId, filename, opts)</code> →
Telemetry Parser	<code>telemetry-parser.ts</code>	CSV parser, FlyWithLua header handling, 1Hz linear-interpolation resampling
Phase Detector	<code>phase-detector.ts</code>	State-machine flight phase classification (taxi, takeoff, climb, downwind, base, final, landing, T&G, stall, steep turn, slow flight)
Constants	<code>constants.ts</code>	V-speeds, KHMP data, arc tolerances, GTGL constants, commitment gate thresholds, geo helpers
Rubrics	<code>rubrics.ts</code>	MetricResult, PhaseResult, SafetyFlag, LessonResult types + scoring utilities
Arc 1 Grader	<code>arc1-grader.ts</code>	Full L1-L4 grading (per-leg scoring, GTGL integration, landing scorer)
Arc Stubs	<code>arc-stubs.ts</code>	L5-L17 stub returning LESSON_NOT_IMPLEMENTED
Landing Scorer	<code>landing-scorer.ts</code>	VS-based severity (Greaser/Smooth/Acceptable/Firm/Rough), weighted composite (VS 50%, hdg 25%, IAS 25%)
GTGL Detector	<code>gtgl-sequence-detector.ts</code>	4-state machine (GA1→TNG→GA2→LAND), builds GTGLResult with pass conditions
Go-Around	<code>maneuvers/go-around.ts</code>	GA detection, commitment gate, early GA detection, safety overrides (stall/touch-down/sink→score=1)
JSON Formatter	<code>json-formatter.ts</code>	

Module	File	Purpose
		<code>toDict()</code> / <code>toJson()</code> matching Python schema, includes extras
HTML Generator	<code>html-generator.ts</code>	Styled debrief HTML with phase breakdowns, coaching bullets
Ground School	<code>ground-school.ts</code>	DB query helper for Ground-SchoolContent
Quiz Helper	<code>quiz.ts</code>	DB query helper for QuizSeed data

Key Constants

Constant	Value	Unit
Vr	55	kts
Vy	74	kts
V_DOWNWIND	90	kts
V_APPROACH	65	kts
V_BEST_GLIDE	68	kts
KHMP Elev	882	ft MSL
Pattern Alt	1,882	ft MSL (1,000 AGL)

Grading Scale

Score	Label	Meaning
1	Rough	Needs significant work
2	Developing	Below standard
3	Solid	Meets standard (passing)
4	Sharp	Above average
5	Nailed it	Excellent

Landing Severity Scale

VS (fpm)	Severity	Color
≤ 100	Greaser	emerald
≤ 200	Smooth	sky
≤ 350	Acceptable	amber
≤ 600	Firm	orange
> 600	Rough	red

L2 Pass Conditions

- GTGL sequence complete (GA1 → T&G → GA2 → Land)
- ≥ 3 pattern cycles completed
- $\geq 2/3$ flap deployment on downwind
- $\geq 2/3$ trim usage in pattern
- Quiz seeds passed (4/5 correct)

4. API Routes

Auth & User

Route	Method	Purpose
/api/signup	POST	Student registration (default STUDENT role)
/api/auth/login	POST	Credential validation
/api/auth/[...nextauth]	*	NextAuth handlers

Lessons & Progress

Route	Method	Purpose
/api/lessons	GET	All lessons with computed status for current user
/api/lessons/[lessonNum]	GET	Single lesson with progress, sessions, quiz, hasQuizSeeds, latestQuizAttempt
/api/sessions	POST	Upload & process telemetry CSV (runs evaluateFlight)
/api/sessions/[sessionId]	GET	Fetch session with scores & attempts (owner/admin/instructor)
/api/progress/debrief-viewed	POST	Mark debrief viewed, evaluate pass conditions (dynamic quiz-aware)
/api/upload/presigned	POST	Generate S3 presigned upload URL

Ground School & Quizzes

Route	Method	Purpose
/api/ground-school/[lesson-Num]	GET	Ground school content + debriefReveal + cfiNotes
/api/quiz-seeds/[lessonNum]	GET	Quiz questions (answers stripped)
/api/quiz-seeds/submit	POST	Submit quiz answers, evaluate pass, update progress
/api/quiz/submit	POST	Legacy check-flight quiz submission
/api/l2-doctrine	GET	L2 GLGL doctrine content

Admin (ADMIN + INSTRUCTOR)

Route	Method	Purpose
/api/admin/students	GET	List all STUDENT users with progress + last 5 sessions
/api/admin/gradebook	GET	Enriched roster: avg score, current lesson, activity status, full sessions
/api/admin/force-complete	POST	Force-complete a lesson for any student (sets score=5, all gates=true)
/api/admin/reset-progress	POST	Delete UserProgress record (revert to LOCKED)
/api/admin/lesson-flags	PATCH	Toggle content readiness flags
/api/admin/shadow-lessons	GET	Fetch lessons with computed status for a target student (impersonation)

5. UI Pages

Page	Route	Key Features
Login/Signup	/login	Email/password auth, framer-motion animations
Dashboard	/dashboard	Path View spine, 17 nodes, arc grouping, shadow mode support, right-click context menu
Lesson Detail	/lesson/[num]	5 tabs: Brief, Ground School, Quiz, Flight, Debrief
Admin Dashboard	/admin	Student list, force complete, readiness flag toggles, gradebook nav
Gradebook	/admin/gradebook	Roster table, search, session drill-down, score breakdowns, HTML debriefs, "View as Student"

Lesson Detail Tabs

Tab	Component	Availability
Brief	brief-tab.tsx	Always (+ L2 Doctrine insert)
Ground School	ground-school-tab.tsx	When ground-SchoolReady=true
Quiz	quiz-seed-tab.tsx	When quizReady=true (Quiz-Seed-based)
Quiz (Legacy)	quiz-tab.tsx	Fallback for check flights with old Quiz model
Flight	flight-tab.tsx	Always (Machado quiz gate for L2/L4)
Debrief	debrief-tab.tsx	After flight uploaded (charts, coaching, GLGL tracker, CFI notes, reveal)

Shadow Mode System

Component	File	Purpose
ShadowContext	lib/shadow-context.tsx	React context: shadowUserId, shadowUserName, isActive
ShadowProvider	lib/shadow-context.tsx	Wraps app via providers.tsx
AdminShadowBar	components/admin-shadow-bar.tsx	Fixed purple banner: “SHADOWING: [Name] EXIT”
providers.tsx	components/providers.tsx	SessionProvider + ShadowProvider + AdminShadowBar

6. User Roles & Access

Role	Dashboard	Lessons	Admin	Gradebook	Shadow Mode
STUDENT	Own path only	Gate-locked	✗ Redirected	✗	✗
INSTRUCTOR	Own path	All accessible	✓	✓	✓ (read-only + overrides)
ADMIN	Own path (bypass)	All accessible	✓	✓	✓ (read-only + overrides)

Security Gates

- **Middleware:** `withAuth` blocks non-authenticated from `/dashboard/*`, `/lesson/*`, `/admin/*`
- **Admin routes:** Server-side role check + API-level role check (ADMIN || INSTRUCTOR)
- **Session detail:** Owner OR ADMIN OR INSTRUCTOR
- **Shadow mode:** Client-side read-only (clicks disabled), context menu for overrides only

7. Gate Logic

Standard Progression

Previous lesson PASSED → Current lesson AVAILABLE

↓ Upload telemetry → flight_score set → IN_PROGRESS or NEEDS_DEBRIEF

↓ View debrief → debrief_viewed = true

↓ Pass quiz (if quizReady) → quiz_passed = true

↓ All conditions met? → PASSED → Next lesson unlocks

Pass Conditions (Dynamic)

- `flightScore >= 3`
- `debriefViewed = true`
- `quizPassed = true` (only if QuizSeed records exist for that lesson)
- L2 additionally: GTGL sequence complete

Admin Override

- Force Complete: Sets `flightScore=5`, `quizPassed=true`, `debriefViewed=true`, `status=PASSED`
 - Reset: Deletes UserProgress record entirely
-

8. Content Status

Lesson Readiness Map

L#	Arc	Title	Flight	Ground	Quiz	Scoring	Media
1	1	Long Walk	✓	✓	✓	✓	✗
2	1	Configur- ator (GLGL)	✓	✓	✓	✓	✗
3	1	Edge of Envelope	✓	✓	✓	✗	✗
4	1	Arc 1 Check	✓	✓	✓	✗	✗
5	2	The Box	✗	✗	✗	✗	✗
6	2	Wind Whisper- er	✗	✗	✗	✗	✗
7	2	Arc 2 Check	✗	✗	✓	✗	✗
8	3	Breaking Free	✗	✗	✗	✗	✗
9	3	Compass Rose	✗	✗	✗	✗	✗
10	3	Arc 3 Check	✗	✗	✓	✗	✗
11	4	Dead Stick	✗	✗	✗	✗	✗
12	4	Weather Wise	✗	✗	✗	✗	✗
13	4	Night Owl	✗	✗	✗	✗	✗
14	4	Arc 4 Check	✗	✗	✓	✗	✗
15	5	Flight Plan	✗	✗	✗	✗	✗
16	5	Dual XC	✗	✗	✗	✗	✗

L#	Arc	Title	Flight	Ground	Quiz	Scoring	Media
17	5	Solo XC	✗	✗	✗	✗	✗

Data Seeding

- `scripts/seed.ts` — 17 lessons, admin user, 4 legacy quizzes
 - `scripts/seed-arc1-content.ts` — L1-L4 ground school + quiz seeds from `data/S2S_Academic_Agent_Master_Syllabus.json`
 - `data/arc1_content_manifest.json` — L1-L4 full content + L5-L17 stubs
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9. Telemetry Parser Field Mappings

CSV Column	Type	Unit	Notes
t_sec	float	seconds	Time since recording start
sim_time	float	seconds	X-Plane sim time
lat, lon	float	degrees	WGS84 coordinates
alt_msl	float	feet	Altitude above mean sea level
alt_agl	float	feet	Altitude above ground level
ias_kts	float	knots	Indicated airspeed
gs_kts	float	knots	Ground speed
vs_fpm	float	ft/min	Vertical speed
pitch_deg, roll_deg, hdg_deg	float	degrees	Attitude
aileron, elevator, rudder	float	-1 to 1	Control deflections
throttle, mixture	float	0 to 1	Engine controls
flap_ratio	float	0 to 1	Flap deployment
on_ground	int	0/1	Weight on wheels
stall_warn	int	0/1	Stall warning active
slip_deg	float	degrees	Ball deflection

Missing columns default to 0. Parser never crashes on missing data.

10. Admin Operations

Login

1. admin@s2s.com / S2SA123
2. All lessons clickable regardless of gate status

Admin Dashboard (/admin)

- Students tab: progress grid (17 dots), force complete buttons
- Lessons tab: readiness flag toggles (5 per lesson)
- Gradebook button → /admin/gradebook

Gradebook (/admin/gradebook)

- Roster: Name, Current Lesson, Avg Score, Last Flight, Status, Actions
- Search: Filter by name or email
- Expand: Click row → session drill-down (score breakdowns, segments, HTML debriefs)
- Shadow: “View as Student” → impersonation mode

Shadow Mode

1. Click “View as Student” in Gradebook
2. Dashboard shows student’s Path View (read-only)
3. Purple banner: SHADOWING: [Name] | Read-Only View | EXIT
4. Right-click any lesson → context menu:
 -  **Pass Lesson** — force complete (score=5, all gates=true)
 -  **Reset Lesson** — delete progress (revert to LOCKED)
5. Click EXIT to return to admin’s own view

11. What Remains

Content Production

- [] L1-L4: Media assets (video content)
- [] L3 scoring: Stall recovery grading
- [] L4 scoring: T&G consistency + emergency scenarios
- [] Arc 2-5 (L5-L17): All content (objectives, briefs, rubrics, ground school, quizzes)
- [] Legacy Quiz 250-253: Expand to full question sets or migrate to QuizSeed

Engine Enhancements

- [] Arc 2-5 graders (currently stubs)
- [] Wind estimation from ground track vs heading
- [] REP walk-around dataref integration
- [] More sophisticated segment boundary detection

Platform Features

- [] Real-time telemetry streaming (WebSocket)
- [] 3D flight replay viewer
- [] Instructor annotation system on debriefs
- [] Student-to-student comparison
- [] Achievement/badge system
- [] Mobile-optimized layout
- [] Instructor account creation UI

12. File Inventory

Scoring Engine (lib/s2s-engine/)

index.ts	☐ Entry point, evaluateFlight()
telemetry-parser.ts	☐ CSV parser, resampling
phase-detector.ts	☐ Flight phase state machine
constants.ts	☐ V-speeds, KHMP, tolerances, geo helpers
rubrics.ts	☐ Types + scoring utilities
arc1-grader.ts	☐ L1-L4 grading logic
arc-stubs.ts	☐ L5-L17 stub grader
landing-scorer.ts	☐ VS-based landing scoring
gtgl-sequence-detector.ts	☐ GTGL 4-state machine
maneuvers/go-around.ts	☐ GA detection, commitment gate, safety overrides
json-formatter.ts	☐ JSON output serialization
html-generator.ts	☐ HTML debrief report
ground-school.ts	☐ DB query helper
quiz.ts	☐ DB query helper

API Routes (app/api/)

auth/login/	☐ Credential login
auth/[...nextauth]/	☐ NextAuth handlers
signup/	☐ Student registration
lessons/	☐ All lessons + computed status
lessons/[lessonNum]/	☐ Single lesson detail
sessions/	☐ Upload + process telemetry
sessions/[sessionId]/	☐ Session detail
progress/debrief-viewed/	☐ Mark debrief viewed
upload/presigned/	☐ S3 presigned URL
ground-school/[lessonNum]/	☐ Ground school content
quiz-seeds/[lessonNum]/	☐ Quiz questions (stripped)
quiz-seeds/submit/	☐ Quiz submission
quiz/submit/	☐ Legacy quiz submission
l2-doctrine/	☐ GLGL doctrine content
admin /students/	☐ Student list
admin /gradebook/	☐ Enriched roster data
admin /force-complete/	☐ Force complete lesson
admin /reset-progress/	☐ Reset lesson progress
admin /lesson-flags/	☐ Toggle readiness flags
admin /shadow-lessons/	☐ Lessons for shadow user

UI Components (app/)

```
login/_components/login-form.tsx
dashboard/_components/dashboard-client.tsx (+ shadow mode + context menu)
lesson/[lessonNum]/_components/
  [ ] lesson-client.tsx      [ ] Tab orchestrator
  [ ] brief-tab.tsx         [ ] Lesson brief (+ L2 Doctrine insert)
  [ ] ground-school-tab.tsx [ ] Ground school content
  [ ] quiz-seed-tab.tsx     [ ] QuizSeed-based quiz
  [ ] quiz-tab.tsx          [ ] Legacy quiz
  [ ] flight-tab.tsx        [ ] CSV upload (+ Machado quiz gate)
  [ ] debrief-tab.tsx       [ ] Debrief (charts, coaching, GLGL, CFI notes, reveal)
  [ ] debrief-charts.tsx    [ ] Recharts visualization
  [ ] glgl-tracker.tsx      [ ] L2 GLGL sequence tracker
  [ ] l2-doctrine.tsx       [ ] L2 go-around doctrine display
admin/_components/admin-client.tsx
admin/gradebook/_components/gradebook-client.tsx
```

Shared Components

```
components/providers.tsx      - SessionProvider + ShadowProvider + AdminShadowBar
components/admin-shadow-bar.tsx - Persistent shadow mode banner
lib/shadow-context.tsx        - Shadow mode React context
```